

CAPE LOOKOUT, NC, USA

WMO: 994160

Lat: 34.622N

Lon: 76.525W

Elev: 5

StdP: 101.27

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -2.4	(c) -0.1	(d) -12.4	(e) 1.3	(f) 0.2	(g) -10.0	(h) 1.6	(i) 2.0	(j) 15.3	(k) 9.5	(l) 14.4	(m) 7.8	(n) 8.0	(o) 350	(p) 0.565

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 3.1	(c) 29.5	(d) 25.9	(e) 29.0	(f) 25.7	(g) 28.5	(h) 25.4	(i) 27.2	(j) 28.6	(k) 26.7	(l) 28.2	(m) 26.3	(n) 27.9	(o) 5.7	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.8	22.5	28.1	26.3	21.7	27.7	25.7	21.0	27.6	85.7	28.5	83.4	28.2	81.5	28.0	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 15.6	(b) 13.4	(c) 11.7	(e) -6.1	(f) 31.2	(g) 2.7	(h) 0.8	(i) -8.0	(j) 31.8	(k) -9.6	(l) 32.2	(m) -11.2	(n) 32.7	(o) -13.1	(p) 33.3		
(4) 15.6	(5) 13.4	(6) 11.7	(7) N/A	(8) N/A	(9) N/A	(10) N/A	(11) N/A	(12) N/A	(13) N/A	(14) N/A	(15) N/A	(16) N/A	(17) N/A	(18) N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	18.0	8.6	9.2	12.4	16.8	20.9	24.8	26.6	26.3	24.3	20.0	14.4	11.2
	DBStd	7.27	5.01	4.13	3.92	3.14	2.64	2.12	1.29	1.47	1.96	3.44	4.14	4.47
	HDD10.0	220	87	58	22	1	0	0	0	0	0	0	11	41
	HDD18.3	1195	303	256	186	63	10	1	0	0	0	24	128	224
	CDD10.0	3138	42	36	96	206	337	446	513	504	431	309	143	77
	CDD18.3	1071	0	0	1	17	88	196	255	246	181	75	10	2
	CDH23.3	7719	0	0	0	4	141	1363	2484	2341	1172	212	2	0
(14) Wind	WSAvg	6.0	6.2	6.6	7.0	6.6	6.0	5.3	5.2	5.3	6.1	5.7	5.9	6.0
	PrecAvg	1423	109	93	96	84	99	103	157	188	191	108	96	110
	PrecMax	2147	216	280	294	225	331	232	344	381	402	301	246	193
	PrecMin	908	44	33	34	18	26	8	65	59	37	3	29	46
	PrecStd	277	48	53	52	53	64	52	63	84	105	71	60	43
	DB	19.7	19.1	21.2	24.0	27.2	29.9	30.2	30.3	28.9	26.9	23.5	21.8	21.8
	MCWB	17.9	17.7	18.1	20.9	24.1	25.2	26.4	26.1	25.0	24.2	21.6	20.7	20.7
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	18.4	17.6	19.8	22.7	25.5	28.9	29.5	29.6	28.1	26.1	22.5	20.3	20.3
	2% MCWB	17.0	16.1	17.7	20.1	23.0	25.2	26.2	25.9	24.7	24.0	20.8	19.1	19.1
	5% DB	17.0	16.3	18.9	21.7	24.8	28.0	29.0	29.0	27.6	25.4	21.5	19.0	19.0
	5% MCWB	15.6	15.0	17.0	19.5	22.4	25.0	25.9	25.6	24.5	23.2	19.6	17.7	17.7
	10% DB	15.7	15.0	17.8	20.7	24.1	27.4	28.5	28.5	27.1	24.7	20.4	17.8	17.8
	10% MCWB	14.3	13.4	16.1	18.6	21.9	24.8	25.6	25.3	24.1	22.3	18.5	16.7	16.7
	WB	19.0	18.0	19.5	21.8	25.0	27.3	27.6	27.9	27.5	26.2	22.3	21.3	21.3
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	MCDB	19.7	18.7	20.2	22.9	26.2	28.1	28.9	29.6	27.5	26.3	23.2	21.9	21.9
	2% WB	17.6	16.8	18.5	21.0	24.0	26.5	27.0	27.4	26.5	25.0	21.4	20.1	20.1
	2% MCDB	18.5	17.2	19.5	22.4	25.3	27.9	28.5	29.0	27.0	25.6	22.4	20.7	20.7
	5% WB	16.4	15.3	17.6	20.3	23.3	25.8	26.7	26.6	25.7	24.0	20.2	18.5	18.5
	5% MCDB	17.5	16.0	18.9	21.5	24.6	27.3	28.3	28.6	26.8	25.2	21.2	19.1	19.1
	10% WB	15.0	13.9	16.5	19.4	22.7	25.3	26.3	26.0	25.2	23.0	18.8	17.2	17.2
	10% MCDB	16.2	15.0	17.8	20.6	23.9	26.8	28.1	28.2	26.7	24.5	20.1	18.0	18.0
(35) Mean Daily Temperature Range	MDBR	6.3	5.9	5.4	4.4	3.6	3.1	3.1	3.2	3.3	4.3	5.5	5.8	5.8
	5% DB	6.9	6.5	5.5	4.4	3.7	3.4	3.4	3.3	3.3	3.7	5.5	6.1	6.1
	5% MCDBR	5.9	6.6	4.4	3.4	2.5	2.4	2.5	2.3	2.9	3.2	5.4	5.9	5.9
	5% WB	6.2	6.3	5.4	4.4	3.4	3.0	3.2	3.4	3.2	3.4	5.4	5.9	5.9
	MCWBR	5.9	6.9	5.0	4.0	2.8	2.5	2.8	2.7	3.0	3.5	5.8	6.0	6.0
(40) Clear-Sky Solar Irradiance	taub	0.330	0.339	0.363	0.401	0.446	0.487	0.512	0.496	0.424	0.374	0.350	0.340	0.340
	taud	2.538	2.511	2.460	2.353	2.273	2.203	2.144	2.185	2.391	2.504	2.519	2.532	2.532
	Ebn at Noon	878	908	910	886	846	808	784	791	837	858	846	843	843
	Edh at Noon	81	93	106	123	135	145	153	144	112	93	83	78	78
(44) All-Sky Solar Radiation	RadAvg	2.55	3.29	4.38	5.56	6.19	6.38	6.00	5.47	4.51	3.84	2.84	2.27	2.27
	RadStd	0.15	0.31	0.33	0.40	0.41	0.36	0.38	0.32	0.29	0.37	0.20	0.16	0.16

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (27 neighbors)	+0.46	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+170	+117

Nomenclature: See separate page